DANIAL AMIN

Data Scientist | Machine Learning Engineer

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Summary

An **Data Scientist** by profession with **experience in dealing with Engineering & IoT Data.** An alumnus of **Georgia Institute of Technology** with **double postgraduate degrees in Aerospace Engineering and Data Analytics.**

An experience of **05 years** in effective **data-driven decision making for maintenance prediction, failure analysis, total quality management, and operational optimization** using platforms including **Python, Google Cloud Platform, IBM Watson, R-Studio, WEKA, and Amazon Web Services ML Platform**.

Experience

Data Scientist

Automat Solutions Inc *02/2022-Ongoing* *Remote*

A world leader in Data Science and Artificial Intelligence.

- ✓ *Developed a Dedicated Pipeline with a series of 14xEstimators with a variety of Pre-processing Strategies for Reliable and Robust Prediction of Battery Properties.*
- ✓ *Selection of effective estimator for effective and efficient prediction of life cycle.*
- ✓ *Conduct of MOO using advanced GE/EA optimization techniques for solution space identification.*

Lead Machine Learning Engineer

Omdena *01/2021-02/2022* *Remote*

A collaborative Artificial Intelligence and Data Science Platform

- ✓ *Working on Reinforcement Learning for detection of receptivity score in twitter audience through twitter and survey based dataset*
- ✓ *Created a model which increased the content reach of non-private organizations for funding and other opportunities by 20%.*
- ✓ *Working on Image Classification and Object Recognition for detection of smuggling activities over sea*
- ✓ *Guided a team of five machine learning engineers with the task of implementation of successful image classification model using CNNs*

Adjunct Faculty Member

IUBH University of Applied Sciences *09/2020-02/2022* *Remote*

A private university offering virtual and distance learning programs in Data Science and Artificial Intelligence.

- ✓ *Trained three cohorts of a class of 40 students in domains of Data Science*

Assistant Manager Aerosapce

NESCOM *08/2017-Ongoing* *Pakistan*

An organization working in the aerospace and defense systems

- ✓ *Created 10+ SOPs in view of Maintenance Data Analysis*
- ✓ *Improved work efficiency by 20% within a year by optimizing Maintenance*
- ✓ *Implemented a Technical Record Management System which optimized the overhauling and maintenance schedules of 52 aircraft components*
- ✓ *Accurately conducted predictive maintenance for 152 aircraft components*
- ✓ *Reduced the risk factor of the autonomous flight by 25% by introducing AI-enabled emergency handling system*
- ✓ *Conducted 23 Root Cause Analysis Sessions with AI Enabled Fault Tracing and Forensic Engineering Mechanism*

Senior Technical Associate

AirSpring Aviation *09/2017-08/2020*

A subsidiary of Bombardier Aviation, working on promotion of new innovative strategies for Business Jet Aircrafts

- ✓ *Developed an optimized and effective way to label the more than 200 aviation accident investigations*
- ✓ *Implemented Fuel Estimation Neural Network Model which reduced the inaccuracy by 500%*

Education

Masters of Science in Advanced Data Analytics
Georgia Institute of Technology
01/2019 - 08/2020 CGPA:4.00/4.00

Masters of Science in Aerospace Engineering
Georgia Institute of Technology
09/2017 - 08/2019 CGPA: 3.95/4.00

Masters of Science in Data Science
IUBH University of Applied Sciences
10/2020 - Ongoing

Micro Masters in Statistics and Data Science
Massachusetts Institute of Technology
05/2020 - Ongoing

Micro Masters in Predictive Analytics using Python
The University of Edinburgh
12/2020 - Ongoing

Micro Masters in Artificial Intelligence

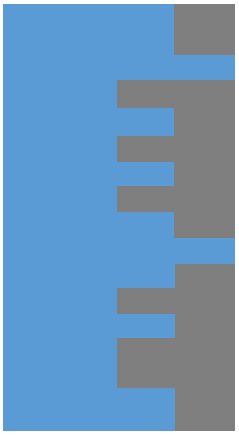
Columbia University
09/2020 - Ongoing

Micro Masters in Design Thinking
Rochester Institute of Technology
08/2020 - Ongoing

Bachelors of Science in Aerospace Engineering
Institute of Space Technology
10/2013 – 08/2017 CGPA: 3.89/4.00

Data Science Capabilities

Data Preprocessing
Data Wrangling
Dimensionality Reduction
Stochastic Optimization
Black Box Optimization
Natural Language Processing
Computer Vision
Sequences and Series
CNNs, RNNs, and DNNs
Data Visualization
Machine Learning
Deep Learning
Reinforcement Learning
Convolutional Neural Networks
Transformers
Gradient Optimization



Core Competencies

Aerospace Data Analysis
Engineering Data Analysis
Foresic Analysis
Root Cause Analysis
Predictive Data Analysis
Stochastic Optimization
Design Optimization

Courses and Trainings

Data Science: Statistics and Machine Learning
A specialization of **05 courses** by **John Hopkins University** on **Coursera**

Data Science
A specialization of **10 courses** by **John Hopkins University** on **Coursera**

Data Science: Foundations using R
A specialization of **05 courses** by **John Hopkins University** on **Coursera**

Executive Data Science
A specialization of **05 courses** by **John Hopkins University** on **Coursera**

Data Analysis and Visualization Foundations
A specialization of **03 courses** by **IBM** on **Coursera**

IBM Data Science
A specialization of **09 courses** by **IBM on Coursera**

Applied Data Science
A specialization of **04 courses** by **IBM** on **Coursera**

Introduction to Data Science
A specialization of **04 courses** by **IBM** on **Coursera**

IBM AI Engineering
A specialization of **06 courses** by **IBM** on **Coursera**

Machine Learning with TensorFlow on Google Cloud Platform
A specialization of **05 courses** by **Google Cloud** on **Coursera**

Python for Everybody
A specialization of **05 courses** by **University of Michigan** on **Coursera**

Deep Learning
A specialization of **05 courses** by **Deeplearning.ai** on **Coursera**

Coding for Designers, Managers, and Entrepreneurs
A specialization of **03 courses** by **University of Virginia** on **Coursera**

Additional Courses
500 other courses from Coursera in the domain of Data Science, 50 Learning Paths (500 courses) from Linked In Learning in domains of Data Science, 100 Courses from Future Learn in domains of Digital Skills Development, 100 Courses from HP E-Learning, 250 Courses from EdX

Projects

Design and Development of Smart Swarm for Multirotor Aircrafts using CNN

Development of Smart and artificially intelligent responsive Flight Control Computer for UAS using Deep Learning Architecture

Development of Effective Flight Data Analysis System using Neural Networks

Adaptive Control System Design for Lateral and Longitudinal Autopilot of a Pilot Less Business Jet Aircraft

Development of Effective Fuel Estimation System using Neural Networks under Machine Learning

Development of SMART AI-enabled Video Capturing and Relay System with Object and Texture Detection

Design and Development of Stock Price Estimator using Prophet Algorithm

Certification

Data Analysis with Python
Developer Certification, representing 400 hours of coursework issued by freecodecamp
Machine Learning with Python
Developer Certification, representing 400 hours of coursework issued by freecodecamp
Data Scientist
Career Path representing 35 weeks of coursework issued by Codecademy
Full Stack Engineer
Career Path representing 6 months of coursework issued by Codecademy
Data Scientist with R
Career Path representing 76 hours of coursework issued by DataCamp
Data Scientist in Python
Career Path representing 102 hours of coursework issued by DataQuest
AI Programming with Pytho
A nanodegree by Udacity

Skills

Programming Languages

Python/ Libraries | Spyder/Jupyter/ Anaconda | R/R-Studio | HTML 5/CSS/JavaScript · C++/C | MATLAB/ Simulink | SQL/MYSQL | LabView | Octave | Numpy/ Pandas/ Tensorflow / SciKit/ Keras/ Matplotlib/ Seaborn

Data Visualization Softwares

Tableau | Power BI | Canva | SPSS | Minitab

Cloud Computing Platforms

Google Cloud | Amazon Web Services | IBM Watson Studio | Microsoft Azure | Google BigQuery

Productivity

MS Office | Zotero/Mendeley | GSuite | Prezi | Latex Google Analytics | Pages | Numbers | Keynote | GitHub

Inter and Intra Personal Skills

Effective Communication Skills | Punctuality & Time Management | Critical Thinking | Analytical Reasoning | Financial Intelligence | Brand Management | Intellectual Property Rights Management | Effective Search and Research Management | Leadership Qualities

Engineering Softwares

CATIA | ANSYS | PRO-E | Dreamviewer | VideoScribe | AutoDesk | Google Sketchup | Simscape | CREO

Publications

Predictive Fuel Estimation of a UAV using Neural Network based Artificially Intelligent Multivariable Predictive Curve Fitting Analysis published in 2019 16th International Bhurban Conference on Applied Sciences and Technology (IBCAST)

Use of Artificial Intelligence-based Flight Data Analysis for Root Cause Identification in Air crashes/accidents published in 2019 16th International Bhurban Conference on Applied Sciences and Technology (IBCAST)

Comparative Analysis of Different Machine Learning Techniques for Development of Flight Data Analysis based RCA Identifier published in 2019 16th International Bhurban Conference on Applied Sciences and Technology (IBCAST)

Languages

English (Proficient) | Urdu (Native)